HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented Python data pipelines for collecting and processing **sensor data** from industrial equipment, aggregating results into a centralized **PostgreSQL database** for analysis.
- Supported the development of **Java Spring Boot** microservices, focusing on building **REST APIs** for inter-service communication and handling core business logic.
- Contributed to the enhancement of an enterprise resource planning (ERP) system by integrating new modules and managing change control processes for system configuration updates.
- Assisted in the instrumentation and execution of **laboratory test programs** for air-cooled chiller components, collecting and reducing **test data** under engineer guidance.
- Supported the selection of mechanical components, including **heat exchangers** and **valves**, by evaluating factors such as cost, performance, and availability for product development.
- Gained understanding of European mechanical and safety codes, including **CE marking** and the **Pressure Equipment Directive (PED)**, by following work instructions and defined processes.
- Containerized Java Spring Boot applications using **Docker** and deployed them to **AWS EC2** instances, managing basic configurations and monitoring.

SKILLS

Core Engineering – Mechanical Design, Fluid Dynamics, Thermal Principles, Refrigeration Systems, European Mechanical Codes, Safety Codes, CE Marking, Pressure Equipment Directive (PED), Machinery Directive, EN 378, EN 14511, F-Gas Regulation, Ecodesign / ErP Framework

Programming Languages – Java, Python, SQL, R

Frameworks & Libraries – Spring Boot, Spring Cloud, pandas, NumPy, scikit-learn, Hibernate, JPA

Databases – PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Statistical Analysis, Data Visualisation

Tools & Methodologies – Jira, Confluence, Agile, Scrum, REST APIs, Linux Shell Scripting

PROJECTS

Industrial Equipment Performance Monitoring System

- Developed a Python-based data ingestion service to collect real-time sensor readings from industrial machinery, processing over **10,000 data points** per hour.
- Designed and implemented a **relational data model** in **PostgreSQL** to store time-series performance data and engineered complex SQL queries for reporting.
- Created interactive **Power BI dashboards** to visualize equipment performance metrics, identify anomalies, and provide operational insights to engineering teams.

Component Selection & Cost Analysis Tool

- Built a Java Spring Boot application that served as a backend for a component selection tool, integrating with a database of available parts.
- Developed RESTful APIs to allow front-end interfaces to query component specifications, costs, and availability, supporting selection for new product designs.
- Collaborated with product managers to define requirements and ensure the tool facilitated efficient evaluation of **heat exchangers**, **fans**, and **valves**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)